Influence Maximization in Social Networks: Classical Algorithms, Learned Estimation, and HIM

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***Abstract*—Influence maximization (IM) selects a small set of users to maximize expected information spread. This review traces the field from greedy optimization to learned influence estimation, with particular attention to HIM's hyperbolic representation and adaptive seed selection. The methods differ in whether they assume a diffusion model, predict spread for whole seed sets, or infer individual scores from cascades. The comparison focuses on how influence is estimated and how overlap among selected seeds is controlled.**

I. Problem Formulation

Given a social graph $G = (V, E)$, a seed budget k, and a diffusion model $\mathcal{M}$, influence maximization seeks

$$S^* \in \arg\max_{S \subseteq V,\ |S| \le k} \sigma_{\mathcal{M}}(S), \qquad \sigma_{\mathcal{M}}(S) = \mathbb{E}[|A_{\mathcal{M}}(S)|], \tag{1}$$

where $A_{\mathcal{M}}(S)$ is the set eventually activated. For a current set S, the marginal gain of adding user v is $\Delta(v \mid S) = \sigma_{\mathcal{M}}(S \cup \{v\}) - \sigma_{\mathcal{M}}(S)$. Since this gain depends on the seeds already chosen, high individual scores need not make the strongest set. Results also depend on $\mathcal{M}$: seeds effective under one assumed process may not be effective under another. The basic objective counts activated users equally; applications with unequal audience value require a different spread measure.

II. From Greedy Guarantees to Scalable Selection

Kempe, Kleinberg, and Tardos (KKT) formalized influence maximization under the independent cascade (IC) and linear threshold (LT) models [1]. In IC, each newly activated user gets one chance to activate each inactive neighbor; in LT, activation depends on the combined influence of active neighbors. Under the standard assumptions of these models, expected spread is monotone and submodular: adding a seed cannot reduce spread, while its marginal gain generally falls as the seed set grows. Greedy selection repeatedly adds the user with the greatest estimated gain and, with exact gains, achieves a $1 - 1/e$ approximation under a cardinality constraint.

The guarantee does not remove the computational bottleneck. Expected spread is usually estimated by repeated simulations, and each greedy round must compare many candidates. CELF uses diminishing returns to reuse previous gain estimates as upper bounds, recalculating a candidate only when it rises to the top of the queue [2]. The cascade example in Fig. 1 illustrates the coverage intuition: a few blogs may capture many observed cascades. Degree discount is a fast local approximation: it adjusts degree-based scores as seeds are selected, using neighborhood overlap as a proxy for redundant reach [3]. It is useful when quick selection matters more than precise spread estimates, but heterogeneous edge probabilities or influence across distinct communities can make degree a poor proxy for reach.

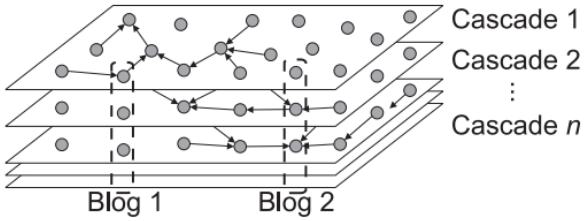


Fig. 1. Information cascades across blogs. Each layer represents a cascade; selecting a few blogs can cover many cascades. Reproduced from Leskovec et al., Fig. 1 [2].

TIM replaces repeated spread simulation with reverse-reachable (RR) sampling. A sampled RR set contains the nodes that can reach a randomly selected target in a sampled live-edge graph; selecting seeds that cover many RR sets estimates spread [4]. This turns estimation into a maximum-coverage problem and supports scalable selection with probabilistic guarantees. CELF preserves the greedy rule, degree discount trades model-based accuracy for a local heuristic, and TIM changes the estimator while retaining a coverage objective. All three assume a specified diffusion process; none infers transmission behavior from incomplete cascades. Later work began learning spread estimates or propagation structure from cascade data, as DeepIM and DeepSN illustrate [5], [7].

III. Learning to Estimate Influence

When diffusion probabilities are unknown, learning-based methods use propagation examples to estimate influence. This shifts effort from repeated inference to training, but the examples still matter: cascades generated by a simulator can transmit that simulator's assumptions to the learned model.

DeepIM learns a latent representation of seed sets, models diffusion patterns, and optimizes seed selection rather than ranking users by fixed individual scores [5]. Figure 2 shows its two-part design: an autoencoder represents the distribution of seed sets, while a learned diffusion model predicts spread; at inference, the latent seed-set representation is iteratively updated. A teacher–student component reduces inference cost, and follow-up work further accelerates this stage [6]. The method can amortize computation across repeated queries, but its advantage depends on training-data cost and whether the

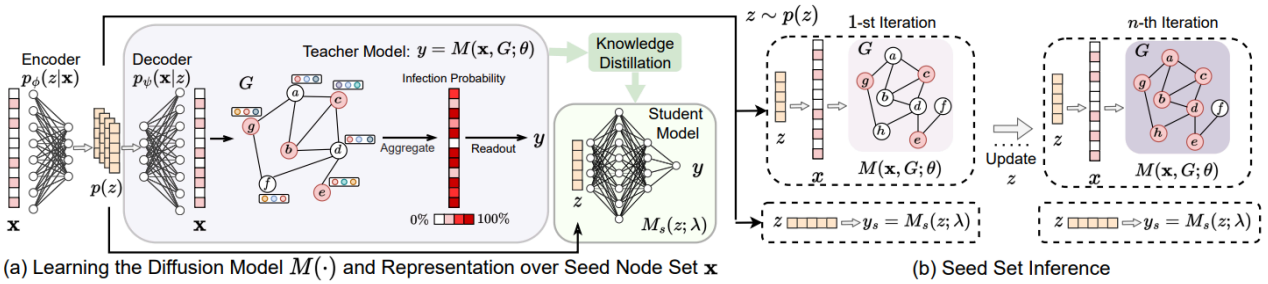


Fig. 2. DeepIM's two-stage framework: learning the diffusion model and seed-set representation (left), followed by iterative seed-set inference (right). Reproduced from Ling et al., Fig. 1 [5].

learned model transfers to new candidate sets or networks. If training cascades come from a simulator, improvements may reflect that simulator's patterns rather than real adoption.

DeepSN takes a propagation-modeling approach, using a sheaf neural framework to represent heterogeneous diffusion and seed-set interactions [7]. Its pipeline first estimates influence with a sheaf GNN and then uses subgraph-guided search to reduce the seed-selection space. This differs from DeepIM's latent seed-set representation and HIM's geometric node estimates followed by a separate selector. DeepIM's search depends on the seed-set distribution represented during training; DeepSN's search depends on which candidates its subgraphs retain. The richer propagation structure may capture interactions a direct predictor misses, but also adds modeling complexity. Their reported gains are not directly comparable unless cascade data, seed budgets, and spread evaluators are aligned.

IV. HIM: Hyperbolic Estimation and Adaptive Selection

Qiao et al.'s HIM addresses settings where the diffusion rule or edge probabilities are unknown but social structure and historical propagation instances are available [8]. Here, model-agnostic means that representation learning and seed selection do not require an explicit IC/LT parameterization. The learned scores still depend on the supplied propagation instances.

Figure 3 summarizes HIM's two stages. Hyperbolic Influence Representation embeds users in a Lorentz-model space, building on hyperbolic methods for hierarchical representations [9], [10]. The rapid growth of hyperbolic space with radius accommodates branching hierarchies, motivating its use for hierarchical propagation patterns. Social edges provide structural information and propagation graphs provide evidence about activation. Rotation-based operations place both signals in the same space; the objective favors observed links, while a regularizer pulls frequent spreaders toward the origin. Negative sampling reduces the cost of estimating edge probabilities. Distance to the origin is used as an individual influence proxy, while pairwise distances encode relations among users. These are learned geometric signals, not direct or causal measurements of adoption. The regularizer gives radial position its influence interpretation; geometry alone does not imply that a user near the origin will reach more people.

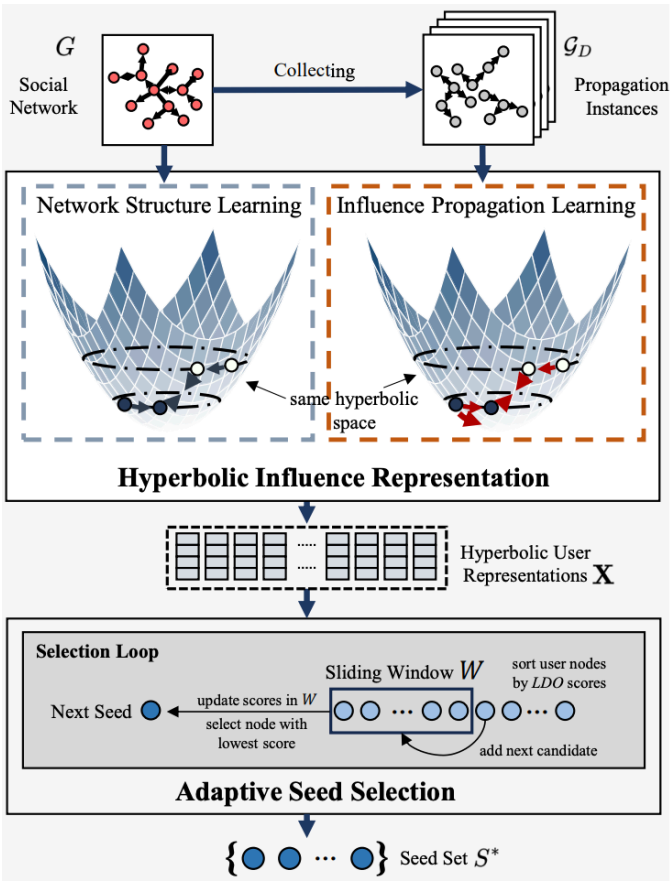


Fig. 3. HIM's learning and seed-selection pipeline. Reproduced from Qiao et al., Fig. 1 [8].

The use of both topology and cascades lets HIM distinguish users with similar degree but different propagation histories. It also makes the representation sensitive to sparse or shifted cascade data. Robustness to a change in cascade distribution requires separate evaluation.

Adaptive Seed Selection (ASW) converts the learned scores into a set. HIM ranks users by LDO, the squared Lorentzian distance to the origin, with lower values indicating greater estimated influence. For embedding $\mathbf{x}_u$ and origin $\mathbf{o}_\mathcal{L}$, $\mathrm{LDO}_u = d^2_\mathcal{L}(\mathbf{x}_u, \mathbf{o}_\mathcal{L})$. The $\mathrm{HIM}_{\mathrm{MD}}$ variant selects the k users with the lowest LDO scores without adaptive updates. ASW starts with the lowest-LDO user and opens a window of size $w = \beta k$.

After each selection, it updates candidates in that window that are also graph neighbors of the chosen seed. The penalty depends on hyperbolic proximity and the selected user's score and degree; the lowest updated candidate is then selected, and the window advances. This local update is cheaper than recomputing network-wide marginal spread, but it may miss overlapping audiences among nonadjacent users. Its behavior depends on both β and whether graph adjacency captures audience overlap. Like Degree Discount, ASW updates candidate scores after each selection, but uses learned geometric proximity to adjust the local redundancy penalty. The paper's $\mathrm{HIM}_{\mathrm{MD}}$ comparison tests this selector against radial ranking using the same representation.

The evaluation covers five networks from Cora-ML to YouTube, with IC/LT settings using known parameters and weighted settings treating parameters as unknown. For example, their Table IV reports Cora-ML WIC coverage of 38.87% for HIM and 38.19% for $\mathrm{HIM}_{\mathrm{MD}}$ at a 10% seed budget [8]. These results evaluate HIM under simulated diffusion, with propagation instances used for training. Simulated cascades on real topologies are not observed opinion change. More tests across budgets, window sizes, cascade volumes, and independently constructed datasets would clarify the limits of the reported gains.

V. Broader Settings, Evaluation, and Outlook

Some extensions change the decision itself. Graph neural bandits use feedback from repeated choices [11], while GBIM uses a graph-based surrogate and Bayesian optimization to select user–information pairs [12]. Temporal models preserve event order; static aggregation can introduce diffusion paths that never existed [13]. THGNets predicts the next activated user with hyperbolic temporal hypergraphs, providing a related representation-learning example [14]. It shares HIM's use of hyperbolic representations, but next-user prediction accuracy does not establish seed-selection performance. These settings differ from the static, user-level seed-selection problem considered by HIM.

A fair comparison aligns the information available to each method, the object being selected, and the spread outcome being measured. It should also report preprocessing and training costs alongside selection time, since learning shifts computation rather than eliminating it. For HIM, useful tests vary cascade volume and recency, transfer across diffusion regimes, and compare ASW with greedy and degree-based selectors under a common evaluator. Where possible, the evaluator should be independent of training labels. Independent training runs measure learning variability; repeated simulations estimate Monte Carlo noise for a fixed model. Both are needed to characterize different sources of uncertainty. Reproducibility requires reporting graph construction, edge direction, cascade generation, probability assignment, and evaluation horizon.

DeepIM predicts spread for candidate seed sets, DeepSN models propagation, and HIM learns geometric user representations before adaptive selection. For HIM, a remaining question is how much overlap its local score updates capture beyond individual radial ranking. Comparisons under a common evaluator can examine this trade-off alongside training and selection costs.

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